

REPORT

BOB-015 — Plugin degenerate-input crash audit

Revision: 1 **Last modified:** 2026-06-09T00:00:00Z **Mandate:** §11.4.118 discovery-pressure (confirm known-issue-set completeness; surface unreported crash-on-degenerate-input gaps before end users hit them) **Method:** read-only enumeration (no source/test edits); throwaway probe run from /tmp, then removed.

Methodology

Reproduced the existing offline-isolation harness pattern from tests/unit/test_plugin_crash_guards.py + tests/unit/test_plugin_parser_guards.py: stub novaprinter (capture prettyPrinter rows) and helpers (retrieve_url, download_file, htmlentitiesdecode, unescape, get_torrent_socks_proxy) into sys.modules, importlib-load each plugin, resolve its class by attribute match (nova2 contract), patch retrieve_url to return each degenerate payload, call search("linux"), and record CRASH vs OK. Run with python3 3.9.x. pytest suite NOT run. No plugins/, tests/, or source files edited.

Probed (15): curated set that ships as files — eztv limetorrents piratebay solidtorrents torlock rutracker rutor kinozal nmclub — plus broad public plugins anilibra bitsearch gamestorrents megapeer nyaa torrentgalaxy torrentkitty. Excluded the 3 already under active repair this session (tokyotoshokan, kickass, yts). The curated names jakeett, torrentproject, torrentscsv do NOT ship as plugin files → out of scope.

Input classes (9): `empty_str`, `whitespace`, `truncated_html`, `malformed_html`, `empty_json {}`, `empty_json_array []`, `malformed_json`, `json_data_null '{"data": null}'`, `none (None)`.

Result matrix (summary)

Every probeable plugin handled all 8 **string** input classes gracefully (0 rows, no exception) — with ONE exception: `piratebay` on `json_data_null`. The `none` input crashed several plugins but is **not reachable** in production (see reachability note).

- All 14 string-probeable plugins: `empty_str` / `whitespace` / `truncated_html` / `malformed_html` / `{}` / `[]` / `malformed_json` → **OK** everywhere.
- `json_data_null` → **OK everywhere EXCEPT piratebay (CRASH)**.
- `none` → CRASH in `eztv`, `limetorrents`, `piratebay`, `solidtorrents`, `torlock`, `bitsearch`, `nyaa`, `torrentgalaxy`; OK in `rutracker`, `kinozal`, `nnmclub`, `anilibra`, `gamestorrents`, `megapeer`, `torrentkitty`.
- `rutracker` → **SKIP-with-reason:** constructor performs a live login (`ValueError: Unable to connect using given credentials.`), cannot be driven offline. Not a crash.

Reachability note (§11.4.6)

`plugins/helpers.py::retrieve_url` is typed `-> str` and every return path returns a `str` ("" on `URLError`; `dataStr` on success) — it NEVER returns `None`. So the `none`-class crashes are **NOT reachable** via the production helpers path; they are defensive observations, not user-facing defects.

NEW reachable guard gap (beyond tokyotoshokan/kickass/yts)

GAP-1 — plugins/piratebay.py crashes on realistic upstream {"data": null}

Reachability: REAL — `retrieve_url` can legitimately return a JSON *object* instead of the expected array (same upstream-shape class `anilibra` was already hardened against in `test_plugin_crash_guards.py`).

Root cause (`piratebay.py search()`): `json.loads('{"data": null}')` → `dict {"data": None}`; the guard `if len(response_json) == 0` sees `len == 1` and does NOT fire; for result in `response_json`: then iterates dict **keys** so `result == "data"` (a str), and `result["info_hash"]` indexes a str with a str.

Verbatim captured evidence (`retrieve_url -> '{"data": null}'`):

```
Traceback (most recent call last):
  File "<stdin>", line 23, in <module>
  File "plugins/piratebay.py", line 91, in search
    if result["info_hash"] ==
"0000000000000000000000000000000000000000000000000000000000000000":
TypeError: string indices must be integers
```

Suggested fix direction: after `json.loads`, if not `isinstance(response_json, list)`: return before the empty-check / iteration — mirroring the existing `anilibra` non-list/null-data guard.

Defensive observations (NOT reachable — no fix needed)

none-only crashers, verbatim messages: `eztv / limetorrents / solidtorrents / torlock / nyaa` → `TypeError: can only concatenate str (not "NoneType") to str`; `bitsearch / torrentgalaxy` → `TypeError: expected string or bytes-like object`; `piratebay (none)` → `TypeError: the JSON object must be str, bytes or bytearray, not NoneType`.

Bottom line

Exactly **one** additional reachable gap beyond the three already fixed: **piratebay on {"data": null}**. No other reachable crash surfaced across 135 (plugin × input) cells. Honest coverage gaps: rutracker (live-login constructor → SKIP) and the un-shipped jackett / torrentproject / torrentscsv curated names were not exercised.